**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 251 · CPU components and registers

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Which data structure supports last-in-first-out Undo?

Q2. What is the extra space complexity of allocating a new length-n array?

Q3. What does the ALU perform?

Q4. Simplify (A ∧ ¬B) ∨ (A ∧ B) and explain why the grouped cells make one variable disappear.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

ALU — Arithmetic Logic Unit.

CU — Control Unit, coordinates and decodes instruction operations.

MAR — Memory Address Register.

MDR — Memory Data Register.

CIR — Current Instruction Register.

PC — Program Counter.

ACC — Accumulator, stores intermediate arithmetic results.

### New learning

Understand the CPU areas — The control unit decodes and coordinates operations; the ALU performs arithmetic and logical operations.

Separate address from data — MAR holds the address being accessed; MDR holds the instruction or data transferred; PC holds the next instruction's address.

Trace the simplified fetch — With PC=120, MAR becomes 120; memory supplies instruction via MDR to CIR; PC advances to the next sequential address. Branching may change PC later.

### Worked example

If PC starts at address 50, trace register roles for the next sequential instruction fetch.

1. Predict first — Copy PC=50 into MAR to identify the memory address being accessed.

2. Explain the answer — Fetch memory content into MDR, copy it to CIR and advance PC to 51.

# We Do · Practise with guidance

If PC initially holds 200 and Memory[200] contains ADD 10, state MAR, CIR and next sequential PC.

```
MAR=200; CIR=ADD 10; PC=201
```

Predict first

MAR must contain 200 and MDR transports the instruction from memory.

Explain the answer

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

CIR now contains ADD 10; for sequential execution PC has advanced to 201.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

State the roles of PC, MAR, MDR, CIR and ACC.

### Question 2

Trace a fetch starting PC=73 where Memory[73] contains LDA 92. State each register's role and the sequential next PC.

### Question 3

Explain why the CPU uses both MAR and MDR rather than a single undifferentiated register, referring to an instruction fetch and a data load.

## Exit Ticket · Answer before checking

The MAR stores the contents of the memory location being fetched.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A stack.

Q2. O(n).

Q3. Arithmetic and logical operations.

Q4. Factor A: A∧(¬B∨B)=A∧1=A. K-map adjacent cells differ only in B, so B is eliminated.

### You Do pages · Independent questions

Q1. [5] PC next instruction address [1], MAR accessed address [1], MDR transferred data/instruction [1], CIR current instruction [1], ACC intermediate arithmetic/logical result [1].

Q2. [6] PC initially 73 [1], MAR becomes 73 [1], memory transfers LDA 92 via MDR [1], CIR receives LDA 92 [1], sequential PC becomes 74 [1], CU interprets the instruction [1].

Q3. [6] MAR carries memory location address [1]; MDR carries actual contents [1]; fetching uses PC→MAR [1] and memory→MDR→CIR [1]; loading data uses an address in MAR [1] and the retrieved value via MDR towards target register/ACC [1].

### Exit Ticket · Model answer

False. MAR holds the address; MDR carries the contents.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding